Villa Complex 415V LV Network

Project summary · REV —

15 October 2026

Owner: Mr. Ahmed Al Mansoori Plot: 123-456 Area: VILLA, UAE Consultant: ABC Engineering Consultants Status: Design

CONNECTED LOAD (TCL)	MAXIMUM DEMAND	TRANSFORMERS	GENERATORS	PANELS
514 kW	**451 kW** 512 kVA · PF 0.88	**1 × 1000 kVA** 51 % loaded · 1 substation(s)	**—**	**5** 1 MDB · 2 SMDB · 1 DB · 1 MCC

TOTAL AREA	POWER DENSITY	CAPACITORS	CABLES	STUDIES
—	**—**	**—**	**1,006 m** 36 cables	**31 to check** 72 pass · 31 warning · 0 fail

Load by type (maximum demand)

HVAC	118 kW	26 %
Motor	76 kW	17 %
Small power / sockets	55 kW	12 %
Fire pump	55 kW	12 %
Lighting	49 kW	11 %
General	47 kW	10 %
EV charging	31 kW	7 %
IT / data	20 kW	4 %

Transformer loading

MDB-1 1000 kVA		51 %

Load per sub-main board

SMDB-GF	203 kW	48 %
MCC-1	130 kW	31 %
SMDB-FF	87 kW	21 %

To do

⚠ **Check** 31 breaker pairs only partly selective

⚠ **Check** MDB-1 PF 0.88 — no capacitor bank yet (100 kvar planned)